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CS 654 MACHINE LEARNING AND DATA MINING

Quick Python Review

WE FLY



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PYTHON FOR MACHINE LEARNING AND DATA MINING

- Simple and easy to learn – a scripting language
- Many machine learning and data mining packages available
- Great for learning and demonstration through interactive mode



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DEVELOPMENT PLATFORM

- Using Anaconda (www.anaconda.com) is encouraged.
- Or install Python (www.python.org) + Jupyter (jupyter.org).
- Our samples will be demonstrated under Jupyter (in a web browser).

FIRST EXAMPLE

- Use print() to display something to the screen:
 - `>>> print("Hello, world.")`
- Which version of Python?
 - `>>> import sys`
 - `>>> print(sys.version)`



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BASIC THINGS

- Comments – use #.
 - `>>> # This is a comment.`
- Data types
 - `>>> a = 5 # a is an integer`
 - `>>> b = 3.14159 # b is a floating point number`
 - `>>> c = 'winter' # c is a string`
- No semicolon.
- No data type declaration.



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TYPE CASTING

- `x = 12`
- `type(x)`
- `str(1024)`
- `int(45.6)`
- `float(98)`



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ARITHMETIC OPERATORS

- `+, -, *, /, %, **`
 - `>>>a = 5+3`
 - `>>>b = 7-9`
 - `>>> c = a * b`
 - `>>> d = 8/3`
 - `>>> r = 7%2` # modulo operation; very useful
 - `>>> p = 3**4` # power operation

COMPARISON

- for numerical values:

- ==, !=, >, >=, <, <=

- for string values:

- ==, !=

LOGICAL CONDITION: TRUE OR FALSE

- if, elif, else

- >>>

```
if (a>b):
```

```
    print('a is greater than b')
```

```
elif (b>a):
```

```
    print('b is great than a')
```

```
else:
```

```
    print('a and b are equal')
```

- indentation, colon, alignment



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STRINGS

- `>>>a = "Hello" # or use 'Hello'`
- `>>>b = "World"`
- `>>> a+b`
- `>>> a+" "+b+'!'`
- `>>> c = a+" "+b+'!'`
- `>>>len(a)`
- `>>>len(c)`



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FUNCTIONS - I

- ```
>>>def greeting():
 print("Hi, there.")
```
- ```
>>>greeting()
```
- ```
>>>def greeting(name):
 print("Hi, " + name + ".")
```
- ```
>>>def yourage(name, age):  
    print("hi, " + name + ". You are " + str(age) + "  
    years old.")
```



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FUNCTIONS - II

- Return your age in 10 years.

```
def age_after_ten(age):  
    return age+10
```

- Return your age before 10 year and after 10 years.

```
def ages_after_before_ten(age):  
    return age+10, age-10
```

MATH FUNCTIONS

- `>>>import math`
- `>>>math.pi;`
- `>>>math.ceil(5.4);`
- `>>>math.sqrt(4);`
- Check Python Math library documentation online.



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FOR AND WHILE LOOPS

- ```
>>>for i in range(10):
 print(i)
```
- ```
>>>for i in range(2,20):  
    print(i)
```
- ```
>>>for i in range(2,20,2):
 print(i)
```
- ```
>>>i = 10
```
- ```
>>>while (i > 0):
 print(i)
 i= i -1
```

# LISTS

- `a = "Hello"`
- `a[0]`
- `b = [45, 21, 19, 33]`
- `len(b)`
- `c = ['red','blue','green']`
- Lists are mutable.
- Combined with for loop.



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# MORE ABOUT LISTS

- Range: `t[:]`, `t[:4]`, `t[2:]`, `t[2:4]`
- `Append()`
- `Insert(i, x)`
- `Sort(reverse=False)`
- multi-dimensional lists
  - `[[...], [...], [.....]]` # a two dimensional list





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# DICTIONARIES

- Key:Value pairs, enclosed by {}
- Create a dictionary:
  - `>>> d = {'one':1, 'two':2, 'three':3}`
  - `>>> d`
  - `{'three': 3, 'two': 2, 'one': 1}`
  - `>>> type(d)`



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# DICTIONARIES CONTD

- Retrieve a value by a key:
  - `>>> d['two']`
- Find out all the keys:
  - `>>> d.keys()`
- Find out all the values:
  - `>>> d.values()`



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# DICTIONARIES CONTD

- Size of the dictionary:
  - `>>> len(d)`
- Add a new key-value pair:
  - `d[newkey] = value`
- Delete a key-value pair:
  - `del d[key]`

# IMPORTS

- Depending on the system setup, we may need to install a package before using it.
- `pip install pandas`
- `import pandas as pd`
- `from pandas import Series, DataFrame`



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# NUMPY ARRAYS

- `import numpy as np`
- `l = [5.2, 1.3, 4.4, 6.7]`
- `a = np.array(l)`
- numpy arrays are often used in applications involving heavy numerical computation.



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